

LOGO REDACTED

Lab Task-02

Department of Electrical and Electronic Engineering

Course Code: EEE-3112

Course Title: Signal and Linear Systems Lab

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Session : Spring-2022

Department : EEE

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Experiment 02: Plotting the Elementary Signals

Objective:

1. Learn the concepts of the elementary signals
2. Plot the elementary signals in MATLAB

Introduction:

To plot a graph, first we need to define two **vectors** corresponding to the independent variable (plotted along x-axis) and the dependent variable (plotted along y-axis). It is worth noting that both the vectors should have the same number of columns. The size of the vectors can be found by calling the **size** function.

Then we call the **stem** function for discrete plot, or the **plot** function for continuous plot. The **stem/plot** function maps the values of the two vectors by their index while plotting.

Lab Task:

Plot the following elementary signals:

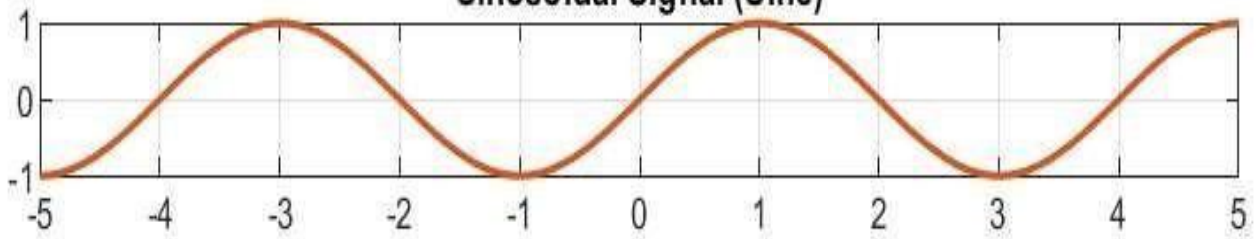
- i. Sinusoidal
- ii. Exponential (damped)
- iii. Impulse
- iv. Step
- v. Ramp

```

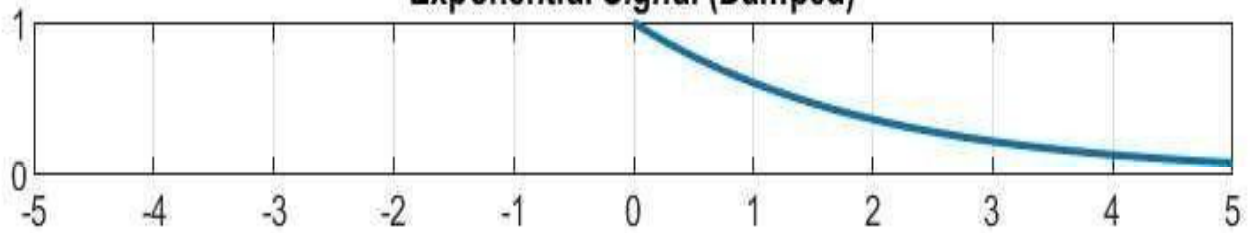
1      % Plotting Elementary Signal
2      % Defining Time
3      t = [-5:0.02:5];
4      % Defining the Signals
5      y1 = sin(2*pi*t/4); % Sinusoidal Signal
6      y2 = exp(-t/2);      % Exponential Signal
7      y3 = t==0;           % Impulse Signal
8      y4 = t>=0;           % Step Signal
9      y5 = t.*y4;          % Ramp Signal
10
11     figure
12     subplot(5,1,1)
13     plot(t,y1,'color','#D95319','linewidth',1.5)
14     title('Sinusoidal Signal (Sine)')
15     xlim([-5,5])
16     grid on
17
18     subplot(5,1,2)
19     plot(t,y2,'color','#0072BD','linewidth',1.5)
20     title('Exponential Signal (Damped)')
21     ylim([0,1])
22     yticks(0:5)
23     xlim([-5,5])
24     grid on
25
26     subplot(5,1,3)
27     plot(t,y3,'color','#7E2F8E','linewidth',1.5)
28     title('Impulse Signal')
29     ylim([0,1])
30     yticks(0:2)
31     xlim([-5,5])
32     grid on
33
34     subplot(5,1,4)
35     plot(t,y4,'color','#4DBEEE','linewidth',1.5)
36     title('Step Signal')
37     ylim([0,1])
38     yticks(0:5)
39     xlim([-5,5])
40     grid on
41
42     subplot(5,1,5)
43     plot(t,y5,'color','#77AC30','linewidth',1.5)
44     title('Ramp Signal')
45     ylim([0,5])
46     yticks(0:5)
47     xlim([-5,5])
48     grid on

```

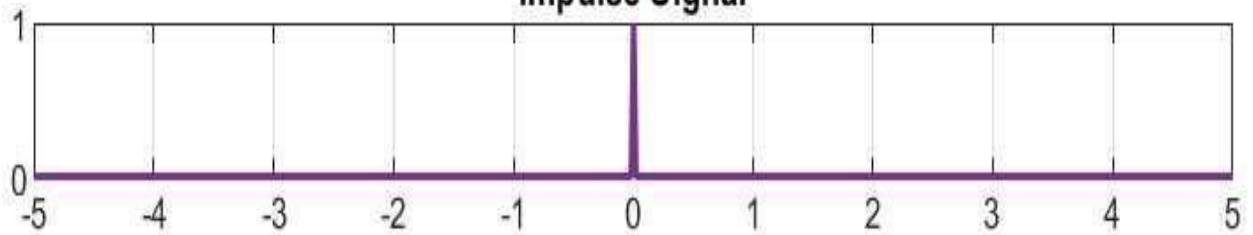
Sinoidal Signal (Sine)



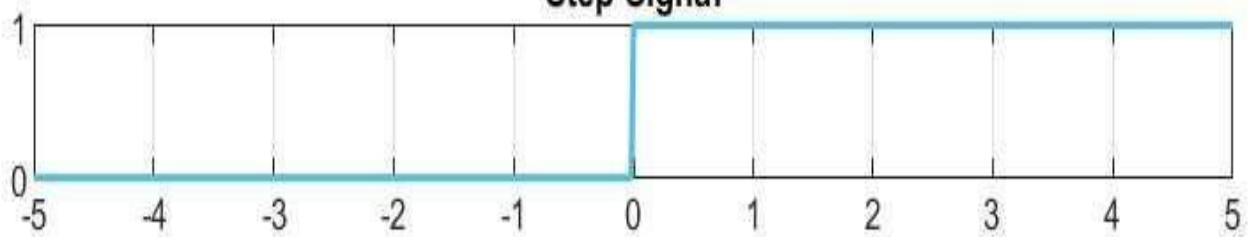
Exponential Signal (Damped)



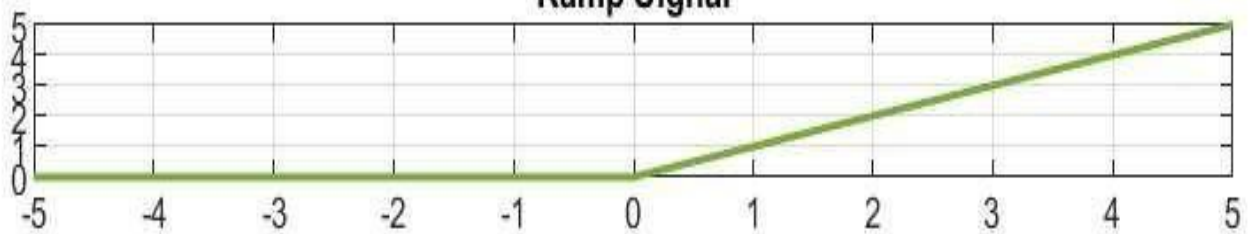
Impulse Signal



Step Signal



Ramp Signal



Lab Exercise:

Plot the following signal:

i. $x(t) = 3 \cdot u(t) + u(t-1) - u(t-2) - u(t-3)$ **Lab Report:**

Do the Lab Task and Lab Exercise.

Include the screenshots of the code and figures in your lab report.

```
1 % x(t) = 3 * u(t) + u(t-1) - u(t-2) - u(t-3)
2
3 - t = -5:0.01:5;
4 - figure
5 - subplot(5,1,1)
6 - x1 = 3*(t>=0);
7 - plot(t,x1,'linewidth',1.5)
8 - title('3*u(t)')
9 - xlim([-5 5])
10 - ylim([0 4])
11 - yticks(0:1:4)
12
13 - subplot(5,1,2)
14 - x2 = (t>=1);
15 - plot(t,x2,'linewidth',1.5)
16 - title('u(t-1)')
17 - xlim([-5 5])
18 - ylim([0 2])
19
20 - subplot(5,1,3)
21 - x3 = (t>=2);
22 - plot(t,x3,'linewidth',1.5)
23 - title('u(t-2)')
24 - xlim([-5 5])
25 - ylim([0 2])
26
27 - subplot(5,1,4)
28 - x4 = (t>=3);
29 - plot(t,x4,'linewidth',1.5)
30 - title('u(t-3)')
31 - xlim([-5 5])
32 - ylim([0 2])
33
34 - subplot(5,1,5)
35 - x = x1+x2-x3-x4;
36 - plot(t,x,'linewidth',1.5)
37 - title('x(t) = 3 * u(t) + u(t-1) - u(t-2) - u(t-3)')
38 - xlim([-5 5])
39 - ylim([0 5])
40 - yticks(0:1:5)
41
```

